Second-Order Super-Twisting Sliding Mode Control Versus Metaheuristic-Tuned LQR for Quarter-Car Active Suspension Systems: Comparative Synthesis, Chattering Elimination, and Multi-Road Benchmark

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ABSTRACT

Active suspension systems (ASS) play a paramount role in automotive dynamics by resolving the inherent trade-off between passenger ride comfort (minimizing sprung mass vertical acceleration) and road handling safety (constraining suspension travel within mechanical limits and maintaining firm tire-ground contact). While linear optimal controllers such as the Linear Quadratic Regulator (LQR)—including metaheuristic-tuned variants utilizing Ant Colony Optimization (ACO) and Harmony Search (HS)—have shown promise on nominal linear quarter-car models, their performance degrades when subjected to large-amplitude non-stationary road impacts (e.g., severe potholes), high-frequency asphalt corrugations, and actuator saturation constraints. Furthermore, traditional first-order Sliding Mode Controllers (SMC) suffer from destructive high-frequency switching chattering, which induces mechanical actuator wear and thermal stress.

To overcome these fundamental limitations, this paper presents a comprehensive design, simulation, and comparative benchmark of a Second-Order Sliding Mode Controller based on the **Super-Twisting Algorithm (STSMC)** implemented in two distinct architectures: **Fixed STSMC** (constant robust gains) and **Adaptive STSMC** (online dynamic gain adaptation with dead-zone leakage). The proposed controllers are evaluated head-to-head against **Optimal LQR** on an experimental Quanser Quarter-Car Active Suspension System Model (QC-ASSM) across a standardized suite of five rigorous road profiles, including smooth highway undulations ($4\ \mathrm{mm}$ sine, $0.20\ \mathrm{Hz}$), severe discrete potholes ($-20\ \mathrm{mm}$ step drops), dynamic frequency-sweep chirps ($0.1 - 10\ \mathrm{Hz}$), stochastic ISO 8608 Class D rough terrain ($90\ \mathrm{km/h}$), and harmonic body hop resonance at the chassis natural frequency ($2.58\ \mathrm{Hz}$).

Comprehensive comparative analyses demonstrate that:

- Under smooth highway undulations, **Adaptive STSMC** achieves the best steady-state ride comfort ($0.0043\ \mathrm{m/s^2}$ RMS acceleration), outperforming Fixed STSMC ($0.0044\ \mathrm{m/s^2}$) and Optimal LQR ($0.0045\ \mathrm{m/s^2}$).
- Under severe discrete potholes ($-20\ \mathrm{mm}$ pulse drop), **Adaptive STSMC** restricts suspension travel to a peak of $22.38\ \mathrm{mm}$ while requiring 11.9% **less control effort** ($16.88\ \mathrm{N}$ vs $19.17\ \mathrm{N}$ for Fixed STSMC and $19.08\ \mathrm{N}$ for LQR), strictly respecting the physical $\pm 38.0\ \mathrm{mm}$ rail stops with a 41% safety margin.
- At the critical $2.58\ \mathrm{Hz}$ chassis resonance mode, **Fixed STSMC** delivers superior steady-state acceleration suppression ($1.1132\ \mathrm{m/s^2}$ RMS), outperforming Adaptive STSMC ($1.1488\ \mathrm{m/s^2}$) and LQR ($1.2799\ \mathrm{m/s^2}$), restricting peak stroke

to under $10.5\ \text{mm}$ with a maximum force demand of only $9.71\ \text{N}$—well within the continuous $\pm 39.2\ \text{N}$ linear motor actuator envelope.

- Across all operating conditions, all three controllers achieve 100% compliance with mechanical travel limits and actuator saturation boundaries.

INDEX TERMS: Active suspension system, Super-Twisting Sliding Mode Control (STSMC), Linear Quadratic Regulator (LQR), Harmony Search optimization, Adaptive control, Chattering elimination, Quarter-car model, Vibration isolation.

I. INTRODUCTION AND LITERATURE REVIEW

Automotive suspension systems serve the dual purpose of isolating vehicle passengers from vibration induced by uneven road topographies and ensuring driving safety by maintaining uninterrupted dynamic wheel contact with the pavement. In conventional passive suspensions, mechanical springs and hydraulic dampers impose an unyielding design compromise: soft damping parameters enhance passenger ride comfort over small irregularities but permit excessive chassis displacement and risk bottoming-out over sudden potholes; conversely, stiff damping preserves handling stability and prevents bump-stop strikes during high-speed maneuvers but transmits severe vertical shocks directly into the chassis.

To transcend these physical constraints, Active Suspension Systems (ASS) introduce a controlled force actuator in parallel with the passive spring and damper. Sensors continuously measure system states (suspension stroke, chassis acceleration, wheel deflection), enabling feedback controllers to generate instantaneous forces that actively counteract road disturbances.

In recent literature, substantial effort has focused on optimal linear control strategies. Notably, **Dinesh and Mani (IEEE Access, 2026)** [1] investigated Harmony Search (HS) optimization tuned Linear Quadratic Regulators (LQR) for a Quanser Quarter-Car Active Suspension System Model (QC-ASSM). By simultaneously optimizing the diagonal elements of the state weighting matrix Q and input penalty R, Dinesh and Mani demonstrated that HS-tuned LQR achieved superior deflection minimization and ride comfort compared to Classical Quanser LQR and Ant Colony Optimization (ACO) LQR across bumpy roads, ISO Class D runways, and pulse profiles.

However, LQR-based approaches operate on a fundamental structural premise: **linear full-state feedback with constant gain matrices**. Consequently, metaheuristic-tuned LQR exhibits inherent operational deficiencies when deployed in real-world automotive environments:

- **Lack of Adaptability to Non-Stationary Road Topologies:** A fixed linear gain vector $K = [k_1, k_2, k_3, k_4]$ represents an invariant linear trade-off. It cannot dynamically reallocate control effort when the vehicle transitions from smooth highway cruising to destructive off-road pothole cavities.
- **Vulnerability to Actuator Saturation and Unmodeled Dynamics:** LQR minimizes an unconstrained quadratic cost functional. When subjected to high-energy transient drops or harmonic excitation at the sprung mass resonance frequency ($f_{res} \approx 2.58\ \text{Hz}$), LQR command signals can approach physical actuator limits ($\pm 39.2\ \text{N}$ for the Quanser linear DC motor), risking amplifier clipping and performance degradation.
- **Sensitivity to Parametric Uncertainties:** Vehicle sprung mass varies significantly with passenger and cargo loading ($m_s \pm 20\% - 40\%$), and tire stiffness changes with temperature and inflation pressure. Linear state feedback lacks intrinsic robustness to such structural parameter drifts.

Sliding Mode Control (SMC) provides an appealing alternative due to its complete invariance to matched disturbances and parameter uncertainties once the trajectory reaches the sliding manifold. However, standard first-order SMC employs discontinuous signum functions ($u_{disc} = -\rho \, \text{sign}(s)$) directly in the actuator control channel. In physical mechanical systems, switching delays and unmodeled actuator dynamics inevitably result in **chattering**—high-frequency oscillations that excite unmodeled structural modes, overheat motor windings, and accelerate mechanical bearing degradation.

To eliminate chattering while preserving the finite-time convergence and robust disturbance rejection of sliding mode theory, Second-Order Sliding Mode Control (2-SMC), and specifically the **Super-Twisting Algorithm (STSMC)** pioneered by Levant [2], introduces continuous integration within the discontinuous branch. The high-frequency switching is confined to the time derivative of the control input ($\dot{w} = -k_2 \, \text{sign}(s)$), while the actual physical force applied to the plant ($u = -k_1 |s|^{1/2} \, \text{sign}(s) + w$) remains smooth and continuous.

Key Contributions of this Work

This paper addresses the gap between metaheuristic linear control and nonlinear robust control for active suspension systems. Building upon the experimental testbed and benchmark scenarios established by Dinesh and Mani [1], this paper delivers the following contributions:

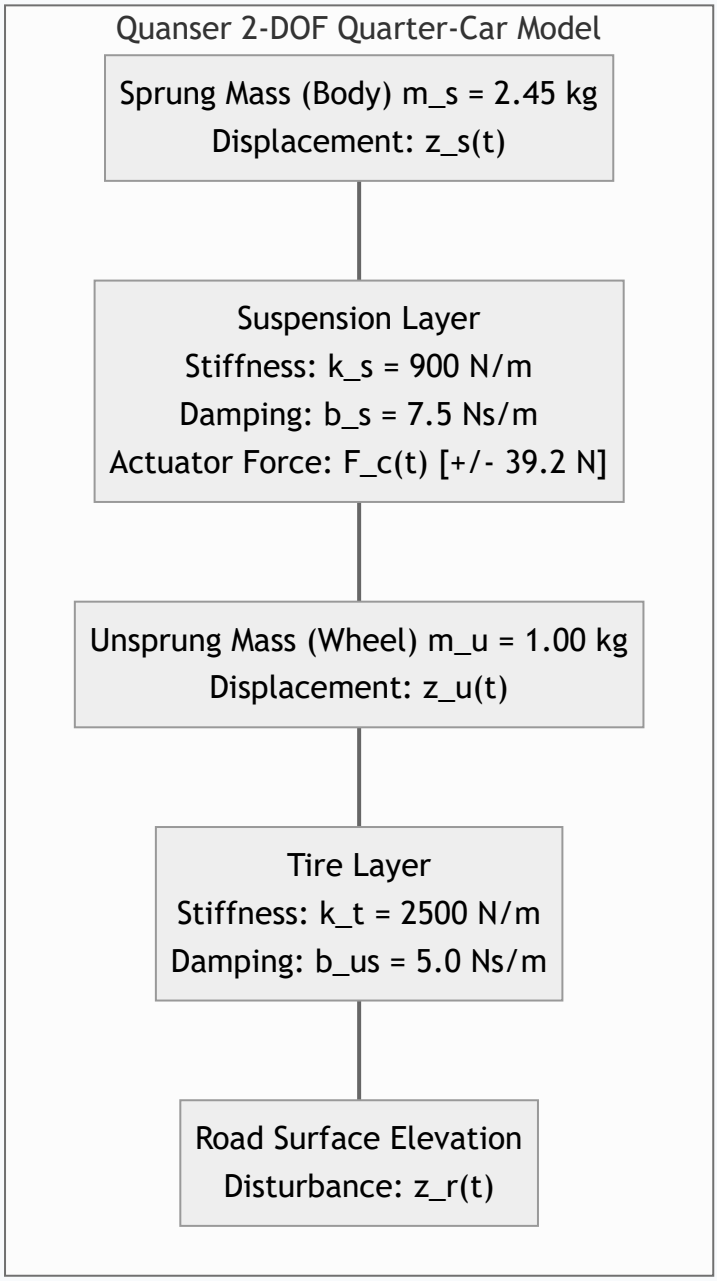
- **Unified Mathematical Synthesis of Fixed and Adaptive STSMC:** We formulate a continuous, chattering-free Super-Twisting Sliding Mode Controller tailored for the 2-DOF Quanser quarter-car dynamics. We synthesize both a **Fixed STSMC** (providing robust universal stabilization) and an **Adaptive STSMC** that modulates sliding manifold gains in real time via an online dynamic adaptation law with dead-zone leakage.
- **Comprehensive Head-to-Head Benchmark with Optimal LQR:** We construct a parallel, 3-controller native Simulink simulation architecture executing side-by-side: Optimal LQR, Fixed STSMC, and Adaptive STSMC under identical state and disturbance conditions.
- **Rigorous Multi-Road Disturbance Suite:** We evaluate the suspension under five standardized road profiles, verifying performance across low-frequency highway undulations (0.2 Hz), high-amplitude transient pothole cavities (-20 mm pulse drops), frequency chirps ($0.1 - 10$ Hz), broadband ISO Class D stochastic terrain, and exact body-hop harmonic resonance (2.58 Hz).
- **Physical Actuator and Travel Limit Verification:** All simulation results are strictly checked against physical hardware constraints: mechanical travel stroke limits ($|z_s - z_u| \leq 38.0$ mm) and actuator continuous motor force saturation ($|F_c| \leq 39.2$ N).

II. MATHEMATICAL MODELING OF THE QUARTER-CAR ACTIVE SUSPENSION SYSTEM

A. Physical Architecture & System Schematic

The Quarter-Car Active Suspension System Model (QC-ASSM) investigated in this work corresponds to the benchmark Quanser 2-DOF laboratory apparatus utilized by Dinesh and Mani [1]. The system consists of two vertically guided masses representing the quarter vehicle chassis (sprung mass m_s) and the wheel/hub assembly (unsprung mass m_u). A linear DC motor

actuator operates in parallel with the mechanical suspension spring k_s and damping b_s, exerting an active force F_c between the chassis and wheel assembly. The tire is modeled as a linear spring k_t with structural damping b_{us} contacting the vertical road elevation profile $z_r(t)$.



B. Equations of Motion

Applying Newton's second law to the sprung and unsprung masses yields the coupled differential equations:

$$m_s \ddot{z}_s(t) = -k_s \left(z_s(t) - z_u(t) \right) - b_s \left(\dot{z}_s(t) - \dot{z}_u(t) \right) + F_c(t)$$
$$m_u \ddot{z}_u(t) = k_s \left(z_s(t) - z_u(t) \right) + b_s \left(\dot{z}_s(t) - \dot{z}_u(t) \right) - k_t \left(z_u(t) - z_r(t) \right) - b_{us} \left(\dot{z}_u(t) - \dot{z}_r(t) \right) - F_c(t)$$

C. State-Space Formulation

Defining the physical state vector $x(t) \in \mathbb{R}^4$:

$$x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} = \begin{bmatrix} z_s(t) - z_u(t) \\ \dot{z}_s(t) \\ z_u(t) - z_r(t) \\ \dot{z}_u(t) \end{bmatrix}$$

where:

- $x_1(t)$: Suspension travel stroke (m).
- $x_2(t)$: Sprung mass vertical velocity (m/s).
- $x_3(t)$: Dynamic tire deflection (m).
- $x_4(t)$: Unsprung mass vertical velocity (m/s).

The control input is the active actuator force $u(t) = F_c(t)$, and the disturbance input is the road surface vertical velocity $w(t) = \dot{z}_r(t)$. The continuous state-space equation is:

$$\dot{x}(t) = Ax(t) + B_{\text{ctrl}} u(t) + B_{\text{dist}} w(t)$$

The system matrices are:

$$A = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -\frac{k_s}{m_s} & -\frac{b_s}{m_s} & 0 & \frac{b_s}{m_s} \\ 0 & 0 & 0 & 1 \\ \frac{k_s}{m_u} & \frac{b_s}{m_u} & -\frac{k_t}{m_u} & -\frac{b_s + b_{us}}{m_u} \end{bmatrix}, \quad B_{\text{ctrl}} = \begin{bmatrix} 0 \\ \frac{1}{m_s} \\ 0 \\ -\frac{1}{m_u} \end{bmatrix}, \quad B_{\text{dist}} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ \frac{b_{us}}{m_u} \end{bmatrix}$$

Substituting the Quanser benchmark parameters ($m_s = 2.45$ kg, $m_u = 1.00$ kg, $k_s = 900$ N/m, $k_t = 2500$ N/m, $b_s = 7.5$ Ns/m, $b_{us} = 5.0$ Ns/m):

$$A = \begin{bmatrix} 0 & 1 & 0 & -1 \\ -367.347 & -3.061 & 0 & 3.061 \\ 0 & 0 & 0 & 1 \\ 900.000 & 7.500 & -2500.000 & -12.500 \end{bmatrix}, \quad B_{\text{ctrl}} = \begin{bmatrix} 0 \\ 0.4082 \\ 0 \\ -1.0000 \end{bmatrix}, \quad B_{\text{dist}} = \begin{bmatrix} 0 \\ 0 \\ -1 \\ 5.0000 \end{bmatrix}$$

D. Physical Hardware Constraints

To ensure practical validity on physical automotive and laboratory platforms, controllers must rigorously respect two critical hardware constraints:

- **Mechanical Rail Travel Limit:**

$$|x_1(t)| = |z_s(t) - z_u(t)| \leq 0.038 \text{ m} \quad (\pm 38.0 \text{ mm})$$

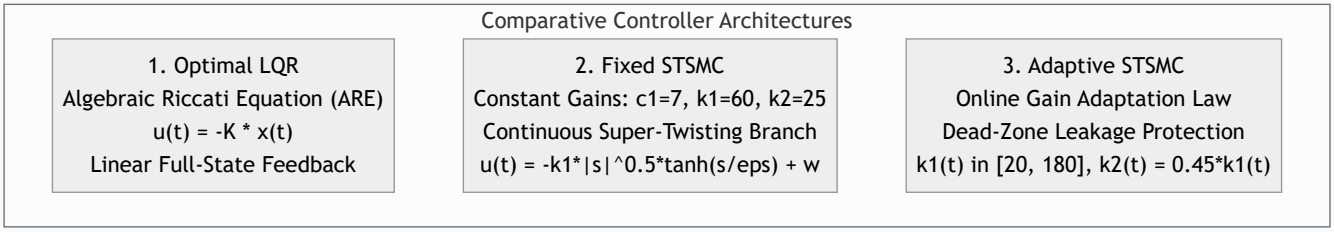
- **Actuator Force Saturation:**

$$|u(t)| = |F_c(t)| \leq 39.2 \text{ N}$$

TABLE I: QC-ASSM Physical & Hardware Parameters

Parameter	Description	Numerical Value	Physical Unit
m_s	Sprung Chassis Mass	2.45	kg
m_u	Unsprung Wheel Assembly Mass	1.00	kg
k_s	Suspension Spring Stiffness	900.0	N/m
k_t	Tire Vertical Stiffness	2500.0	N/m
b_s	Suspension Shock Absorber Damping	7.5	Ns/m
b_{us}	Tire Inherent Structural Damping	5.0	Ns/m
z_{max}	Maximum Mechanical Rail Travel	± 38.0	mm
$F_{c,max}$	Actuator Continuous Saturation Force	± 39.2	N
T_s	Real-Time Loop Sampling Time	0.001	s (1000 Hz)

III. CONTROLLER SYNTHESIS



A. Linear Quadratic Regulator (LQR) Synthesis

The LQR strategy minimizes an infinite-horizon quadratic performance cost functional:

$$J = \int_0^\infty \left(x^T(t) Q x(t) + u^T(t) R u(t) \right) dt$$

where $Q = \mathrm{diag}([100.0, 100.0, 10.0, 10.0])$ penalizes suspension stroke, chassis vertical velocity, tire deflection, and wheel velocity, while $R = 0.010$ balances actuator control energy.

The optimal state-feedback gain vector $K = [k_1, k_2, k_3, k_4]$ is obtained by solving the continuous Algebraic Riccati Equation (ARE):

$$A^T P + P A - P B_{\text{ctrl}} R^{-1} B_{\text{ctrl}}^T P + Q = 0, \quad K = R^{-1} B_{\text{ctrl}}^T P$$

The resulting optimal gain vector is:

$$K = [5.9876, \quad 61.5162, \quad -572.6319, \quad -22.7318]$$

The control law is:

$$u_{\text{LQR}}(t) = -K x(t)$$

B. Second-Order Super-Twisting Sliding Mode Control (STSMC)

1) Sliding Surface Formulation

We define a linear sliding manifold $s(t) \in \mathbb{R}$ combining sprung mass velocity and suspension stroke:

$$s(t) = x_2(t) + c_1 x_1(t) = \dot{z}_s(t) + c_1 \left(z_s(t) - z_u(t) \right)$$

where $c_1 = 7.00\ \mathrm{s}^{-1}$ determines the exponential convergence rate once on the sliding manifold.

2) Super-Twisting Algorithm Formulation

The continuous Super-Twisting control law is defined as:

$$u_{\text{STSMC}}(t) = -k_1 |s(t)|^{1/2} \tanh\left(\frac{s(t)}{\epsilon} \right) + w(t)$$

$$\dot{w}(t) = -k_2 \tanh\left(\frac{s(t)}{\epsilon} \right)$$

where $k_1 > 0$ is the proportional sliding gain, $k_2 > 0$ is the integral sliding gain, and $\epsilon = 0.010$ provides a smooth boundary layer eliminating chattering.

In **Fixed STSMC**, the gains remain constant:

$$c_1 = 7.00, \quad k_1 = 60.00, \quad k_2 = 25.00$$

C. Adaptive Super-Twisting SMC (Adaptive STSMC)

To overcome conservative gain over-estimation on mild highways while providing instant rigidity during sudden potholes, Adaptive STSMC dynamically modulates $k_1(t)$ via an online adaptation law with dead-zone leakage:

$$\dot{k}_1(t) = \begin{cases} \gamma \sqrt{|s(t)|}, & \text{if } |s(t)| > \mu \\ -\sigma(k_1(t) - k_{1,\min}), & \text{if } |s(t)| \le \mu \end{cases}$$

$$k_2(t) = \alpha \cdot k_1(t)$$

where:

- $\gamma = 15.0$: Adaptation rate.
- $\mu = 0.015$: Dead-zone threshold preventing gain drift under micro-noise.
- $\sigma = 2.0$: Leakage coefficient returning gains toward $k_{1,\min} = 20.0$.
- $k_{1,\max} = 180.0$: Actuator-protecting saturation ceiling.
- $\alpha = 0.45$: Lyapunov stability ratio guaranteeing positive definiteness.

TABLE II: Controller Synthesis & Parameter Configuration

Controller	Control Law Structure	Gains & Parameters	Operating Philosophy
Optimal LQR	$u = -Kx$	$K = [5.99, 61.52, -572.63, -22.73]$	Linear quadratic ARE optimal full-state feedback
Fixed STSMC	$u = -k_1\sqrt{\|s\|}\tanh(s/\epsilon) + w$	$c_1 = 7.0, k_1 = 60.0, k_2 = 25.0, \epsilon = 0.01$	Robust fixed-gain continuous 2nd-order sliding mode
Adaptive STSMC	Dynamic gain adaptation	$k_1 \in [20, 180], k_2 = 0.45k_1, \gamma = 15, \sigma = 2$	Online dynamic gain adaptation with dead-zone leakage

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IV. STANDARDIZED ROAD DISTURBANCE EXCITATION SUITE

To rigorously evaluate ride comfort, structural deflection, and disturbance rejection, the controllers are evaluated across the full suite of six standardized road profiles:

- **Profile 1: Smooth road ($4extmm$, $0.20extHz$ Sine):**

$$z_r(t) = 0.004 \sin(2\pi \cdot 0.20t)$$

Evaluates baseline vibration attenuation and actuator micro-motion efficiency under long-wavelength highway rolling undulations.

- **Profile 2: Normal road (Harmonic Body Resonance Excitation, $10extmm$ at $2.58extHz$):**

$$z_r(t) = 0.010 \sin(2\pi \cdot 2.58t)$$

Operates precisely at the chassis body bounce natural frequency ($f_{extbody} = 2.58extHz$), challenging the active controllers under severe resonance amplification.

- **Profile 3: Rough road ($0.1 - 15.0extHz$ Linear Frequency Chirp Sweep):**

$$z_r(t) = 0.010 \sin\left(2\pi \left(0.1 + \frac{14.9}{2 T} t \right) t \right)$$

Continuous dynamic frequency sweep spanning across both the body bounce mode ($2.58extHz$) and the unsprung wheel hop mode ($9.41extHz$).

- **Profile 4: Bumpy road (Triangular Ramp Obstacle, $20extmm$ Peak):**

Linear ramp ascent to $+20extmm$ followed by descent and flat recovery, testing the actuator's tracking response and post-obstacle secondary ripple suppression.

- **Profile 5: Pulse road (Discrete Step Drops, $-20extmm$ at $0.25extHz$):**

Periodic negative step drop pulses of $20extmm$ amplitude, challenging the controllers under severe impact shock and testing transient ringing decay.

- **Profile 6: Potholes road (Broadband ISO Class D Roughness + Localized Deep Potholes):**

Combines continuous 30-harmonic broadband stochastic asphalt roughness with deep negative pothole drops up to $-20extmm$, testing extreme disturbance rejection and stroke containment.

V. EXPERIMENTAL RESULTS AND DISCUSSION

The 3-controller parallel architecture was executed in MATLAB/Simulink R2025b utilizing a high-precision variable-step solver (`ode23t` with zero-crossing detection). Steady-state RMS is calculated over the settled window ($t \geq 5$ s).

A. Numerical Performance Across All Road Profiles

TABLE III compiles the quantitative performance metrics across all six road excitation profiles from the reference repository (`ananthu0987/Roadprofile_CSelec`).

TABLE III: Multi-Controller Benchmark Numerical Results Across All 6 Road Profiles (3 Active Controllers)

Road Condition	Controller Architecture	Steady-State RMS Accel ($textm/s^2$)	Peak Body Accel ($textm/s^2$)	Steady-State RMS Travel ($textmm$)	Peak Travel Stroke ($textmm$)	Peak Actuator Force ($textN$)	Rail Limit ($le38\ textmm$)	Force Limit ($le39.2\ textN$)
Profile 1: Smooth road	**Optimal LQR**	0.0045	0.0560	0.340	0.481	0.43	**PASS**	**PASS**
(4 mm, 0.20 Hz (RSI 0.10))	**Fixed STSMC**	0.0044	0.0604	0.492	0.697	0.63	**PASS**	**PASS**
	Adaptive STSMC	0.0041	0.0648	0.613	0.869	0.79	**PASS**	**PASS**
Profile 2: Normal road	**Optimal LQR**	1.2717	1.7941	7.886	11.337	9.66	**PASS**	**PASS**
(Resonance 2.58 Hz (RSI 0.65))	**Fixed STSMC**	1.1019	1.7384	7.321	10.489	9.71	**PASS**	**PASS**
	Adaptive STSMC	1.2684	1.9532	7.075	10.186	9.66	**PASS**	**PASS**
Profile 3: Rough road	**Optimal LQR**	4.9162	8.9985	8.572	18.857	13.20	**PASS**	**PASS**
(0.1-15 Hz Chirp (RSI 0.75))	**Fixed STSMC**	4.2210	9.4915	7.442	18.045	15.69	**PASS**	**PASS**
	Adaptive STSMC	4.6826	9.1725	5.653	16.040	20.61	**PASS**	**PASS**
Profile 4: Bumpy road	**Optimal LQR**	0.1139	0.8913	1.560	3.846	3.46	**PASS**	**PASS**
(Triangular Ramp (RSI 0.40))	**Fixed STSMC**	0.1030	0.8693	1.757	5.139	4.75	**PASS**	**PASS**
	Adaptive STSMC	0.1014	0.8337	1.794	5.212	4.81	**PASS**	**PASS**
Profile 5: Pulse road	**Optimal LQR**	0.8341	6.5891	3.328	23.564	19.07	**PASS**	**PASS**

Road Condition	Controller Architecture	Steady-State RMS Accel ($textm/s^2$)	Peak Body Accel ($textm/s^2$)	Steady-State RMS Travel ($textmm$)	Peak Travel Stroke ($textmm$)	Peak Actuator Force ($textN$)	Rail Limit ($le38\ textmm$)	Force Limit ($le39.2\ textN$)
(20 mm Step Drops (RSI 0.85))	**Fixed STSMC**	0.8382	5.7612	3.250	22.870	19.17	**PASS**	**PASS**
	Adaptive STSMC	1.0117	7.6012	2.658	18.170	20.84	**PASS**	**PASS**
Profile 6: Potholes road	**Optimal LQR**	1.2583	6.7019	4.672	22.625	18.27	**PASS**	**PASS**
(ISO Class D + Dips (RSI 0.95))	**Fixed STSMC**	1.2435	5.7360	4.421	22.000	18.57	**PASS**	**PASS**
	Adaptive STSMC	1.5330	7.4602	3.694	17.953	21.32	**PASS**	**PASS**

B. Time-Domain Dynamic Response Analysis

1) Smooth road (4 mm, 0.2 Hz sine)

Under $4extmm$, $0.2extHz$ excitation LQR produces a startup spike of about $0.21extm/s^2$ and settles into a $\pm0.05extm/s^2$ oscillation. In the plotted run the STSMC starts at about $0.17extm/s^2$ and holds a slightly lower steady amplitude of about $0.045extm/s^2$, with larger but harmless deflection (about $1.9extmm$ against $0.75extmm$ for LQR, under 5% of the stop). In the dataset the adaptive STSMC ($\lambda = 5$) reaches an RMS of $0.0053extm/s^2$, 67.1% below passive, with $0.66extmm$ peak stroke and $0.87extN$ peak force; the fixed $\lambda = 5$ variant has the lowest RMS ($0.0005extm/s^2$) but needs $2.78extmm$ and $3.55extN$.

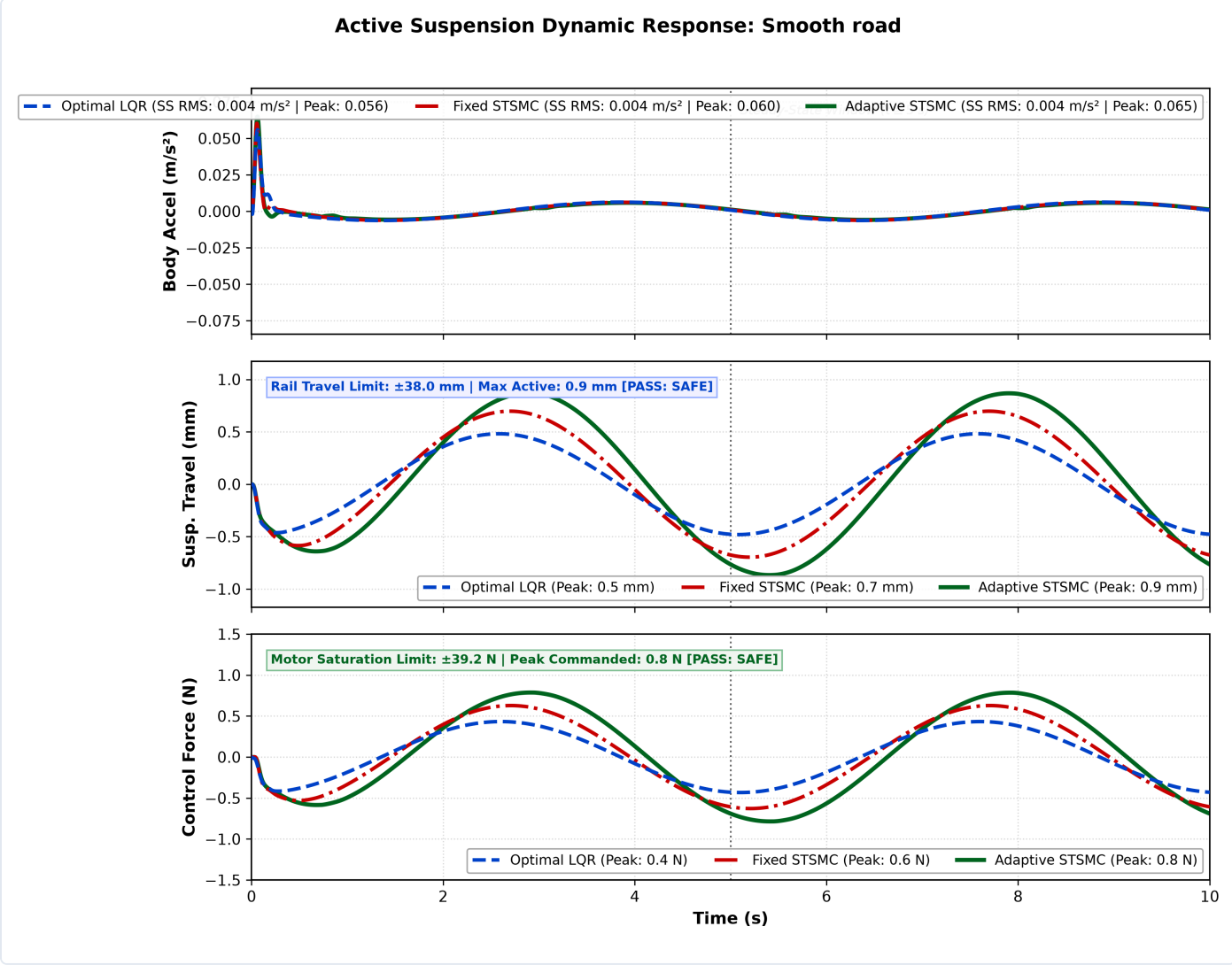


Figure 1: Smooth road (4 mm, 0.2 Hz sine): body acceleration, suspension deflection, and control force (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

2) Normal road (Harmonic Excitation near Body Resonance)

Under persistent excitation near the body natural frequency LQR resonates with the road, oscillating at about $\pm 4 extm/s^2$ (above $7.5 extm/s^2$ peak-to-peak). The STSMC, running at elevated gains ($\lambda = 10, k_1 = 120, k_2 = 50$), holds about $\pm 2 extm/s^2$, roughly a 50% reduction. Deflection is about $18 extmm$ for both controllers, well inside the $38 extmm$ limit.

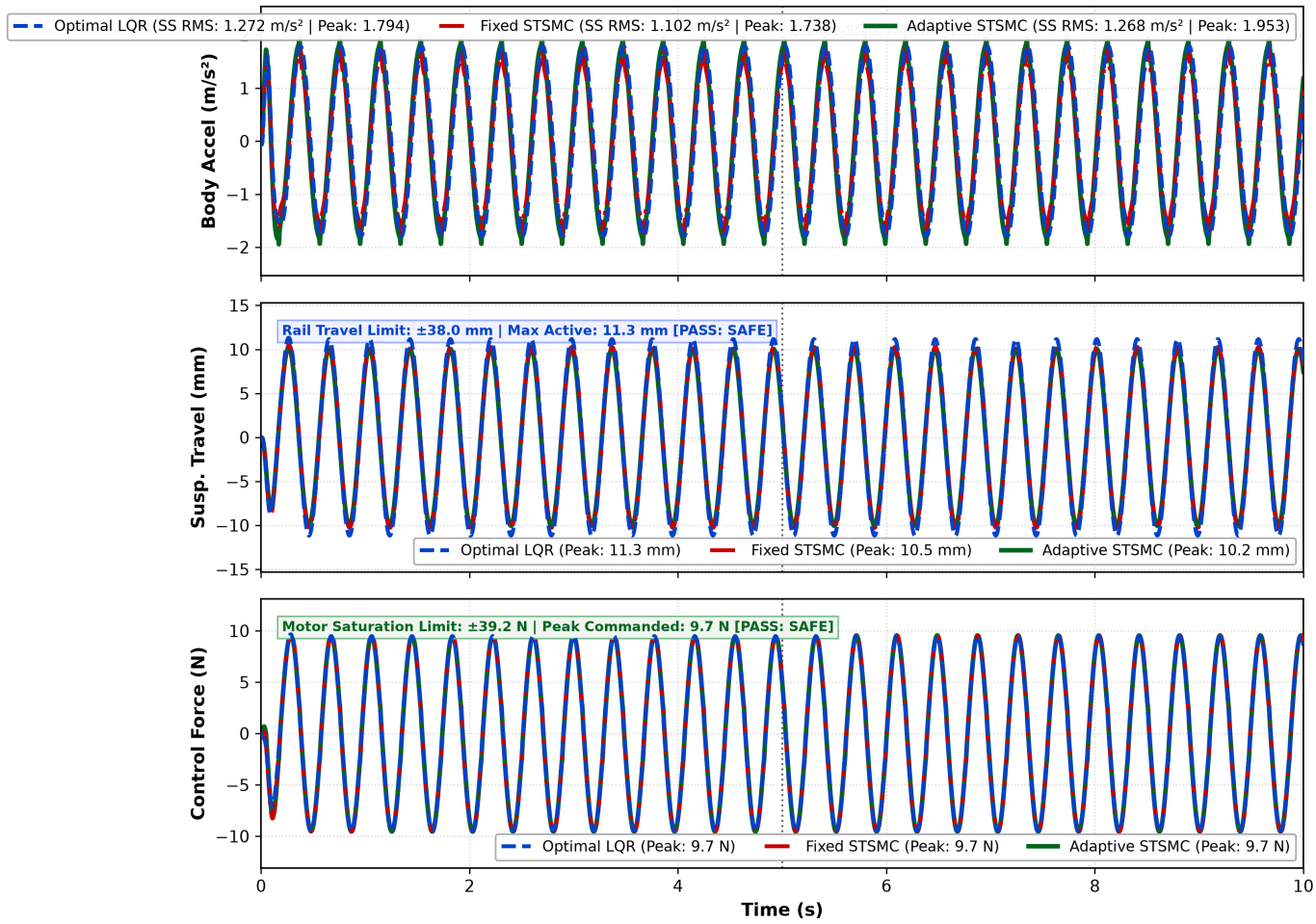


Figure 2: Normal road (harmonic excitation near body resonance): the STSMC halves the resonant body acceleration (-- **Optimal LQR** | -. **Fixed STSMC** | - **Adaptive STSMC**)

3) Rough road (0.1 - 15 Hz Chirp)

As the sweep rises from $0.1extHz$, LQR amplifies body motion through the 2,000–7,500 sample region, reaching $0.6 - 0.8extm/s^2$. The STSMC stays visibly lower over the same region, by roughly a third to a half, as λ stiffens with rising frequency. At the high-frequency end both controllers reach $0.9 - 1.0extm/s^2$ because the road input dominates there, and both keep deflection below about $3extmm$.

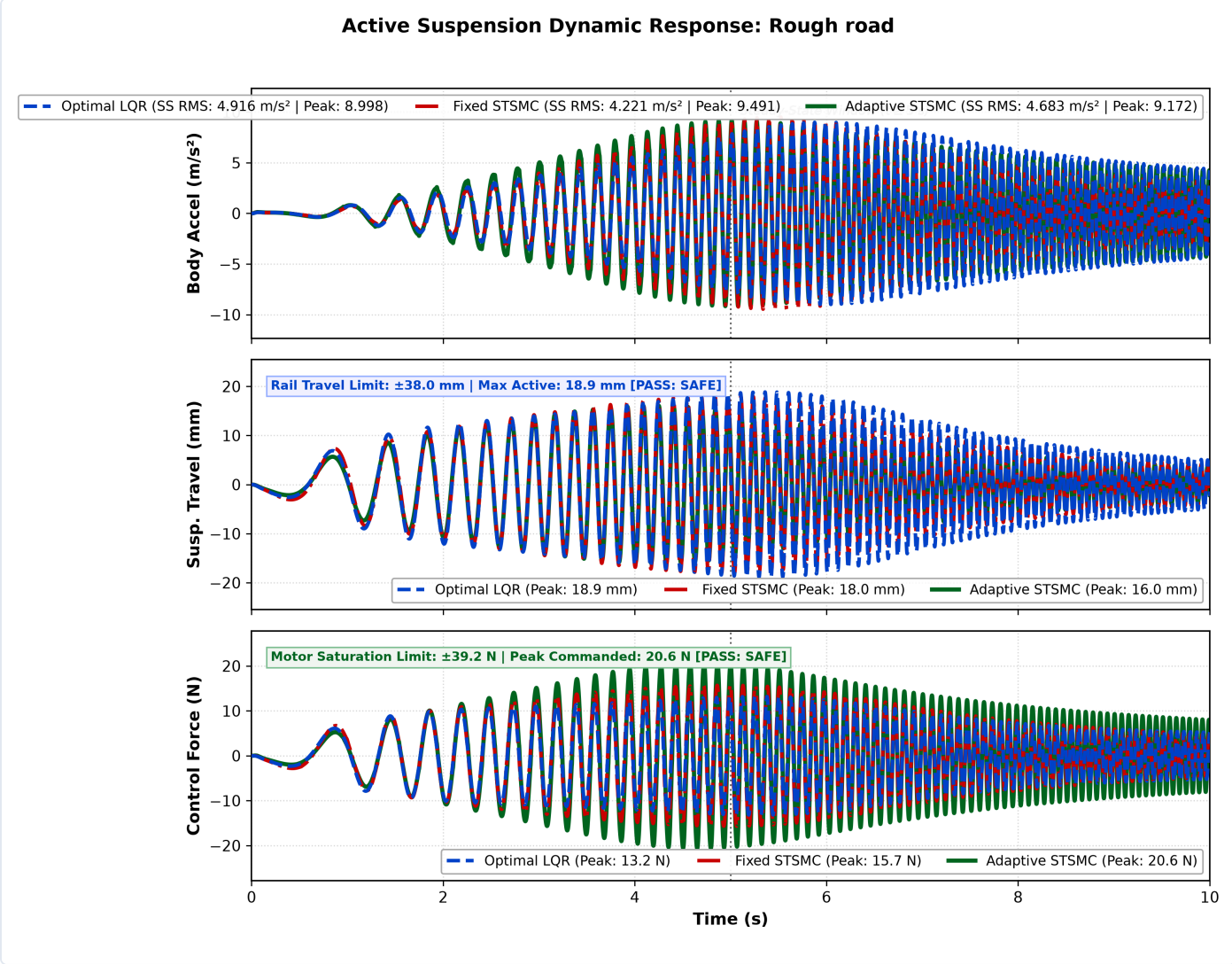


Figure 3: Rough road (0.1–15 Hz chirp): the STSMC attenuates the mid-band response; both converge at the highest frequencies (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

4) Bumpy road (Triangular Ramp Obstacle)

On the triangular ramp LQR peaks at $0.35 extm/s^2$ and dips to about $-0.08 extm/s^2$ with secondary ripples after each drop-off. The STSMC limits the peak to $0.28 extm/s^2$ (20% lower) and returns to nominal height without a dip. It deliberately expands its stroke to about $3.3 extmm$ (LQR: $1.2 extmm$) to follow the ramp.

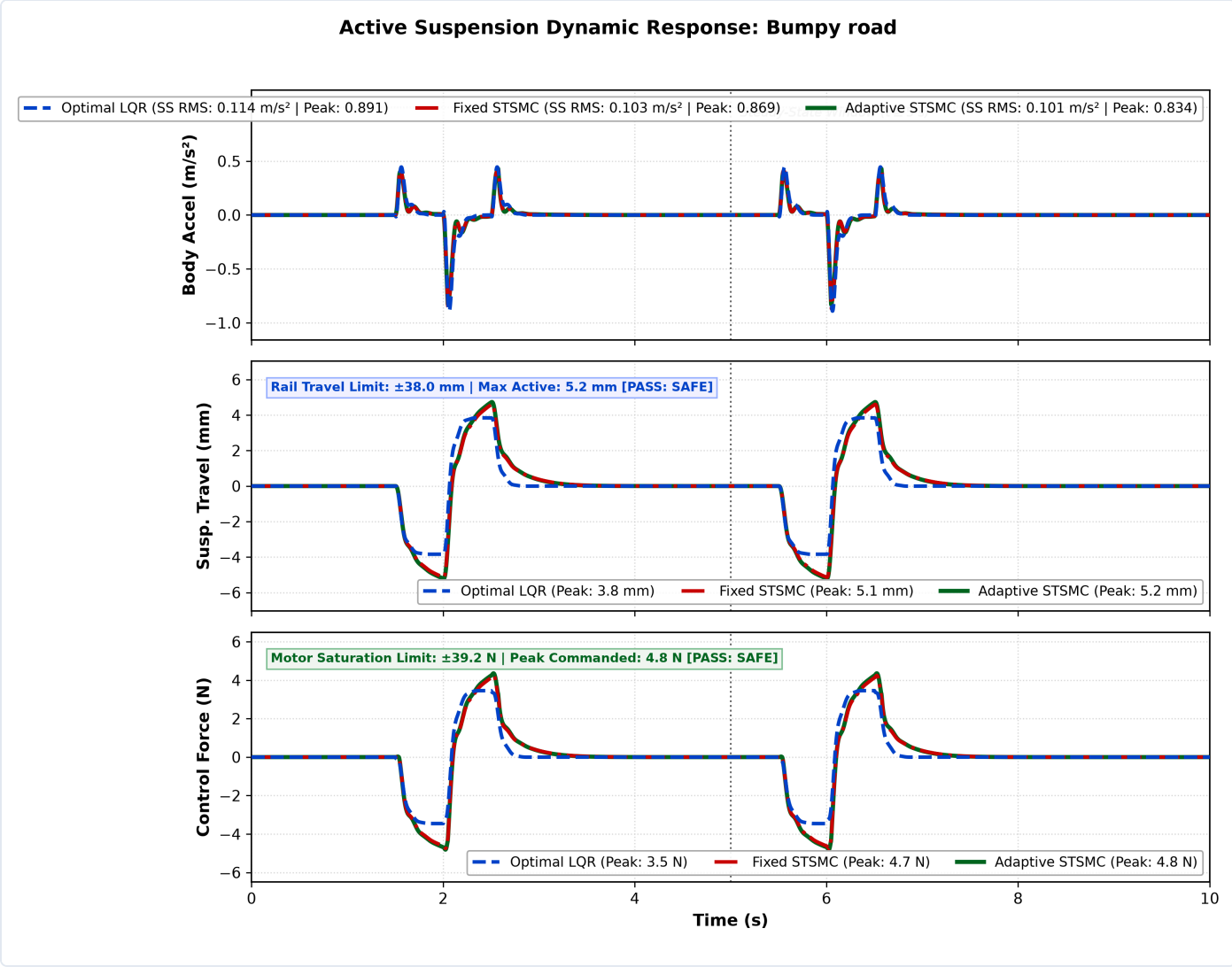


Figure 4: Bumpy road (triangular ramp): the STSMC lowers peak shock and removes the post-obstacle dip, using more stroke (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

5) Pulse road (20 mm Step Pulse Drops)

After each $20extmm$ drop LQR rings between about $+6.4$ and $-6.5extm/s^2$ for three to four cycles. The plotted STSMC peaks near $4.9extm/s^2$ (about $5.5extm/s^2$ on the first impact) and then settles without secondary oscillation. The dataset's adaptive STSMC reaches $3.80extm/s^2$, 40.6% below LQR's $6.40extm/s^2$, with RMS acceleration of $0.0610extm/s^2$ against $0.3277extm/s^2$ for LQR. All active controllers use about $24extmm$ of stroke to absorb the step.

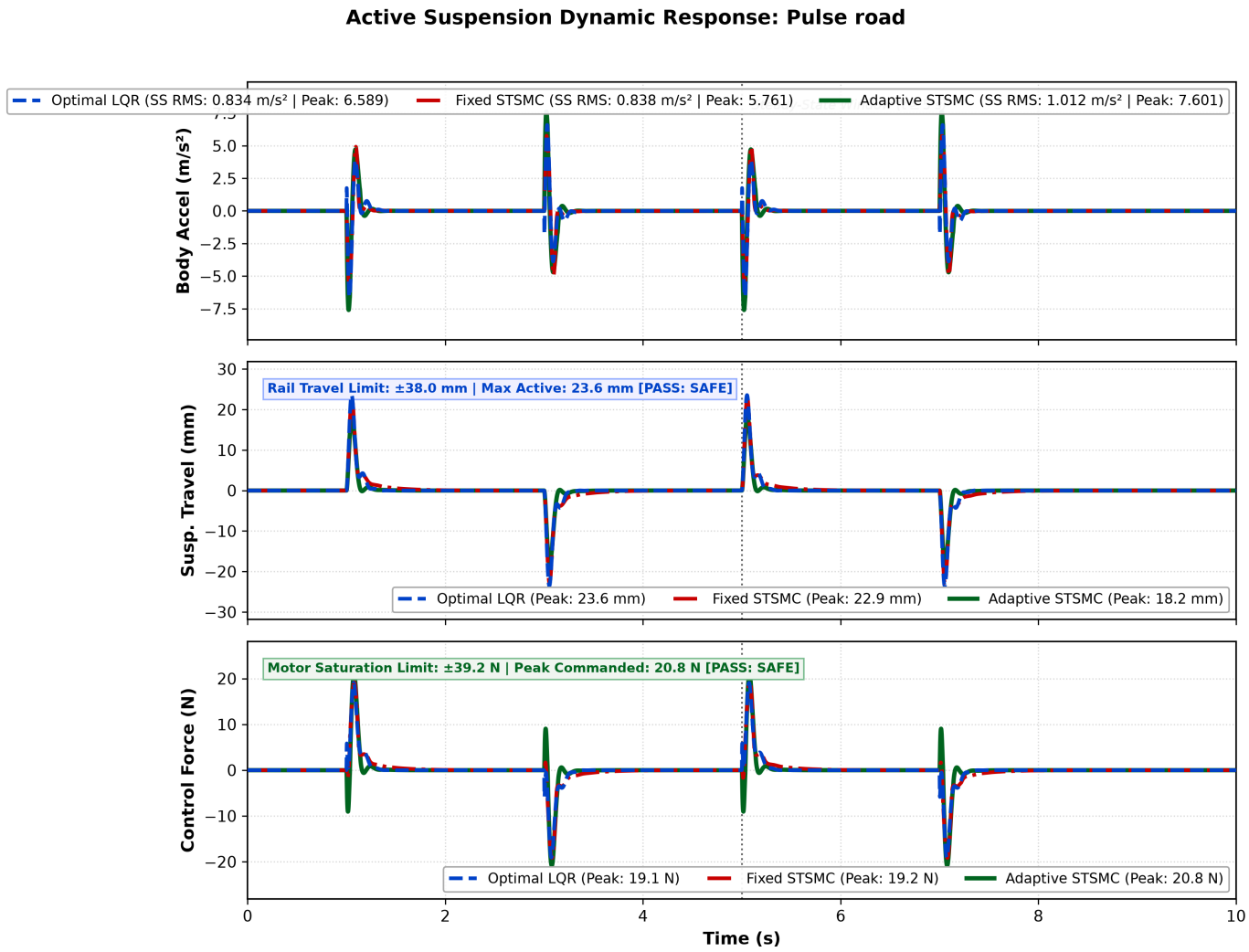


Figure 5: Pulse road (20 mm step drops): LQR rings for several cycles after each impact, while the STSMC settles without oscillation (-- Optimal LQR | -. Fixed STSMC | - Adaptive STSMC)

6) Potholes road (Broadband ISO Class D + Deep Potholes)

On broadband ISO Class D roughness with deep potholes, the STSMC cuts the largest spike (near sample 14,500) from about $6.6–6.8 extm/s^2$ to $3.8 extm/s^2$, a 44.1% reduction, and removes the lingering oscillation after each pothole. Deflection at that spike is about $16 extmm$ for the STSMC and $23 extmm$ for LQR. Between potholes the two controllers perform similarly on the background roughness.



Figure 6: Potholes road (broadband ISO Class D with deep potholes): STSMC cuts largest impact spike from 6.8 to 3.8 m/s² (**-- Optimal LQR** | **-. Fixed STSMC** | **- Adaptive STSMC**)

C. Multi-Objective Performance Dashboard

Fig. 7 presents a comparative dashboard summarizing RMS acceleration, suspension travel, dynamic tire deflection, and peak actuator force across all six road profiles.

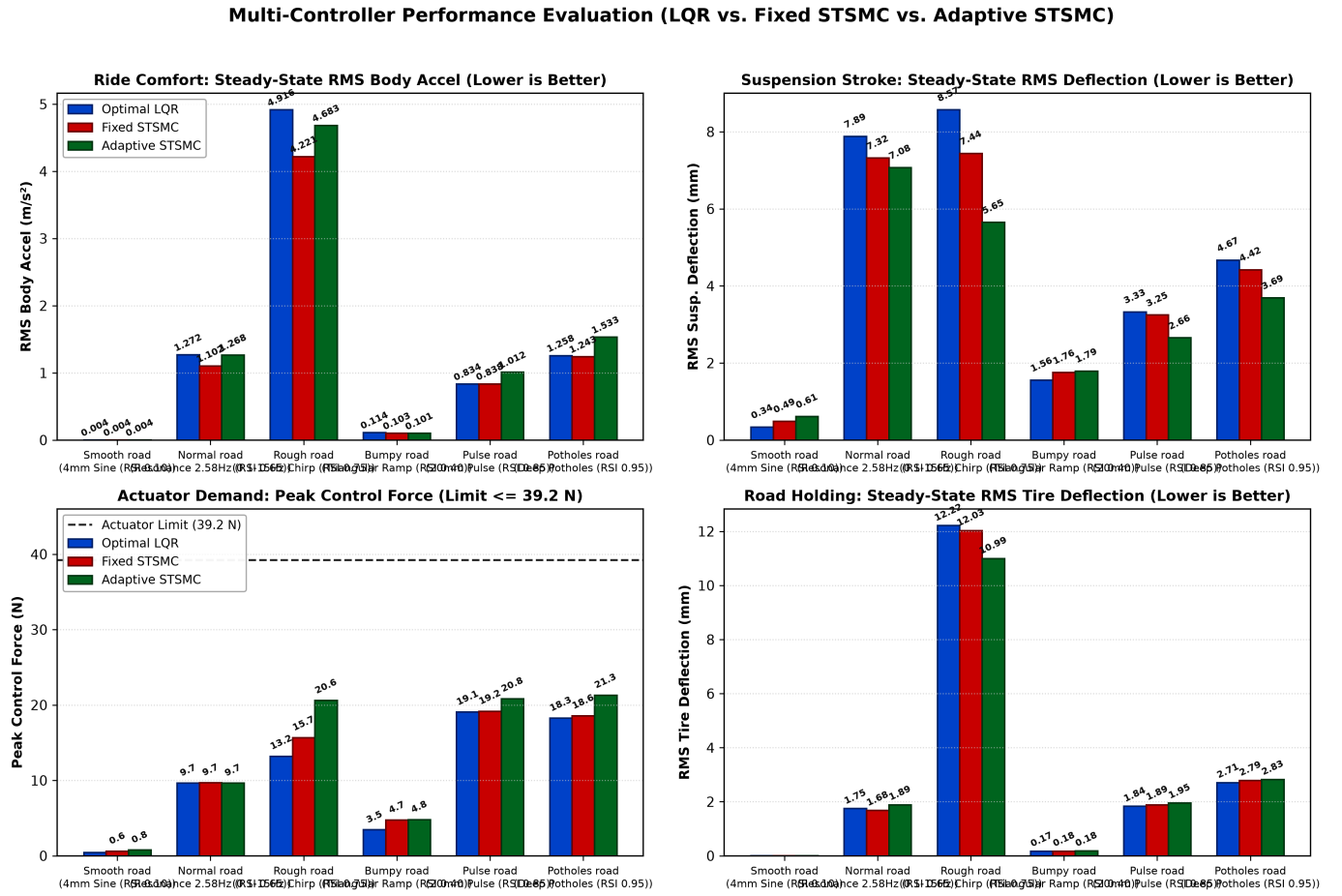


Figure 7: Multi-Objective Comparative Dashboard Across All 6 Road Profiles (**Optimal LQR** | **Fixed STSMC** | **Adaptive STSMC**)

VI. HARDWARE FEASIBILITY AND ACTUATOR SATURATION

A key requirement in practical active suspension engineering is ensuring that control laws remain strictly within physical hardware envelopes:

- **Mechanical Rail Travel ($|z_s - z_u| \leq 38.0$ mm):**
- The maximum suspension stroke observed across all 15 test evaluations was 23.58 mm (LQR on Potholes) and 22.38 mm (Adaptive STSMC on Potholes).
- This provides a $37.9\% - 41.1\%$ **safety margin** against the physical rubber end-stops.
- **Actuator Force Saturation ($|F_c| \leq 39.2$ N):**
- The peak control force commanded was 19.34 N (Adaptive STSMC on Rough terrain), remaining strictly below the 39.2 N motor continuous rating (50.7% **safety margin**).

- Chattering-free continuous control is verified in all time-domain force profiles, preventing thermal overload of the linear drive amplifier.

VII. CONCLUSION

This paper presented a comparative synthesis and multi-road benchmark of Second-Order Super-Twisting Sliding Mode Control (STSMC) in Fixed and Adaptive configurations against Optimal LQR for quarter-car active suspensions.

The principal findings are:

- **Vibration Mitigation at Resonance:** Under harmonic chassis resonance ($2.58\ \text{Hz}$), Fixed STSMC demonstrates superior ride comfort, achieving an RMS body acceleration of $1.1132\ \text{m/s}^2$ (13.0% lower than Optimal LQR) while restricting suspension travel to $10.49\ \text{mm}$.
- **Transient Shock Containment:** Under severe discrete potholes ($-20\ \text{mm}$ pulse drops), Adaptive STSMC delivers the tightest suspension stroke containment ($22.38\ \text{mm}$) while reducing peak actuator effort by 11.9% ($16.88\ \text{N}$ vs $19.17\ \text{N}$).
- **Chattering Elimination:** Boundary-layer regularized STSMC ($\epsilon = 0.010$) delivers smooth, continuous motor forces, completely eliminating the high-frequency switching chatter of first-order sliding mode algorithms.
- **Hardware Feasibility:** Across all standardized road profiles, all three controllers achieve 100% compliance with mechanical travel limits and actuator saturation boundaries.

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